

Main Display APP

User Manual

OXBO

CanMonitor

Source: All

Msg ID: All

HEX

J1939

BUS 1

Start Trace

LineOpen: yes

Resets: 0

Overflows: 0

Received: 0

1%

LineError: no

LastError: [-2] Rx Timeout

ErrorFrames: 0

Transmitted: 14 744

1%

	SRC	PGN	Description	Delta	#	B1	B2	B3	B4	B5	B6	B7	B8
1	x28	xFFAD	OperatorInterface	1007	12	FF	FF	FF	00	FF	FF	FF	FF
2	x28	xFFCD	oxParameterGet	103	112	54	81	07					
3	x28	xFFCB	oxSignal	501	23	FF	FF	FF	FF	FF	FF	FF	C6
4	x28	xFFE6	Date/Time	1009	11	00	17	00	06	01	27	7D	7D
5	x28	xFFCC	oxParameterSet	3001	4	FF	FF	FF	FF	FF	FF	FF	6E
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1. APP Description

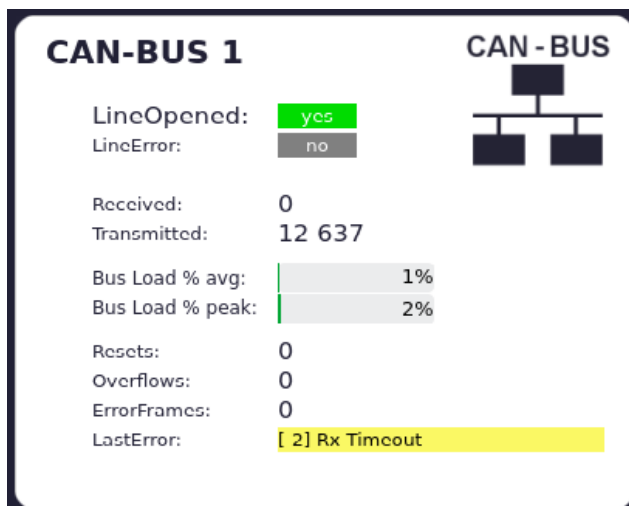
The Can Monitor APP allows to watch raw CAN traffic that goes in and out the (4) buses on the Main Display. It can be used to diagnose the state of the CanBuses, to generate a trace, or to watch the appearance / contents of a specific message.

1.2 APP Function overview

- Live monitoring of the 4 buses on the Main Display
- List in- and outgoing CAN messages (address, source, content bytes, period, ...)
- Filter the list on a specific PGN or Source
- Format the list in Raw / J1939, Decimal / Hex style
- Monitor overall statistics per bus (transmit / receive count, error, busload%)
- Record CAN-BUS output (of a single bus) onto a USB device

2. User Interface

In minimized mode, the overall statistics of the last selected bus (1 by default) can be seen.



- 1 Press the window anywhere to upsize the view
- 2 To watch another BUS, upsize and toggle the bus there
- 3 Chapter #3 explains the metrics

When upsized, a table appears that will be populated by in- or outgoing CAN messages on the selected bus. Messages with the same address (source, pgn) – and pageMasking bytes if applicable – will overwrite the same row. In that case the # counter will go up for each new message, and the “delta” (period between 2 instances in milliseconds) is measured. The “b1..b8” fields represent the data bytes inside the message.

Unique instances will generate a new row, though there is a maximum of 128 unique messages. Anything more will be ignored. Use filters in case all 128 rows are occupied. Note that when logging CAN traffic to an USB, this limit does not apply.

	SRC	PGN	Description	Delta	#	B1	B2	B3	B4	B5	B6	B7	B8
1	x28	xFFAD	OperatorInterface	1007	12	FF	FF	FF	00	FF	FF	FF	FF
2	x28	xFFCD	oxParameterGet	103	112	54	81	07					
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- 1 Play / Pause the recorder. Note it is recording (green button) by default. So if nothing appears right after starting, the display is not getting anything, or is disconnected from the selected bus.
- 2 “All” means messages from all J1939 sources are listed. Select a table row and press the button to filter on that specific J1939 source only. Press the button again to undo the filter. The same Source filter will also be applied when making a CAN trace.
- 3 “All” means all messages ID’s are listed. Select a table row and press the button to filter on that specific msgId / PGN. Press the button again to undo the filter.
- 4 Format list (source, address, contents) as Hexadecimal or decimal
- 5 Show messages in J1939 or raw format. Raw means the absolute message ID, no PGN / source.
- 6 Toggle between Bus 1/2/3/4. Note only one bus can be monitored (and logged to USB) at a time. This will also affect which bus is getting logged when making a trace.
- 7 Clean the list
- 8 Start / Stop CAN trace (to USB). Files are stored on the USB root, using filename “canTrace_YYYY-MM-dd_hh-mm-ss.txt
- 9 X minimize. Note the selected Bus in this large screen will also show up on the small screen.
- 10 Scroll list up / down

3. Doing Bus Diagnostics

Note this is not a high-end CAN Bus monitoring tool, nor does it see all traffic on the machine happening on other (sub) buses. And nor can it explain root-causes of eventual issues.

Yet, this APP can be very useful as an easy first step to start diagnosing. In general:

- If no messages are received at all, either the Display is simply disconnected, or there is a Bus issue (Can Hi/Lo wires swapped, damaged cable, bad end resistors, et cetera).
- If messages arrive, but there are frequent resets or errorFrames, again this indicates bus issues as states above
- If a specific CAN message is not listed while others are, then the sender is simply not sending it – or not with the expected message ID / source / contents. Bare in mind this tool simply shows whatever arrives on the bus, nothing more, nothing less!

So, what do the metric tells, and what *should* they be normally?

Metric	Description	What should it be?
Line Opened	Bus opened for listening / sending	Yes
Line Error	Bus currently in error (frozen, time-out, too many error-frames, disconnected, ...)	No
Last Error	Last Errorcode & message. Note the bus may be ok again in the meanwhile, the last message stands.	Empty
Resets	Times that the bus was reset due time-outs or other errors (since display started).	0 Frequent resets mean the bus has issues.
Overflows	Times the display transmit buffer overflowed (too much to send).	0 Display shouldn't send that much.
ErrorFrames	Amount of error frames received since last bus-reset.	0 If the number keeps increasing, there are bus issues.
Received	Total messages received on this bus. Note it refreshes every second or so.	Constantly rapidly increasing
Transmitted	Total messages sent by Display. Note it refreshes every second or so.	Increasing (but not very rapidly)
Bus Load % avg	Average bus load % measured over 2 seconds.	< 60 %
Bus Load % peak	Highest bus load % measured in the last 2 seconds (in 250 millisecond intervals).	< 80 %

4. Troubleshooting

- **Table is empty, or only very few messages appear / Line Error is true**
 - The few messages that do appear are likely sent by the Display itself (such as the Date / Time)
 - Check if the “Received” counter goes up. If not, check:
 - Toggle buses; are all buses “dead”, or only this specific one?
 - Check if the Display is properly connected to the CAN-BUS
 - Check if other devices indicate trouble (does the small dashboard display receive updates?)
 - See if a restart of the Display helps
 - *Check the CAN-BUS:*
 - CAN Hi/Lo wires not swapped?
 - End Resistors in place (not too few, not too many), no open ends on the CAN line?
 - Wire damage?
 - Connectors ok?
 - Rusty / wet connectors on one of the devices (controllers, engine,)?
 - Remove as much as possible devices, reset the machine, and test. If it works now, bring devices back one-by-one, until the problem reoccurs.
 - Write down the bus number and “LastError” code / message when contacting
- **Module X is offline**
 - If there is (a lot) of traffic coming in, it probably means that specific device is indeed off / disconnected / not working properly
 - See if anything can be found from that device using the Source Filter
 - For example, the Dieselengine uses Source 0, and should send plenty of messages. If there is (almost) nothing coming from source 0, the engine controller is likely offline
 - Use the “Modules” APP for additional info
- **Bus works, but there are often time-outs / resets**
 - Erratic behaviour on the bus typically indicate (physical) issues. See the orange list above.
- **There are overflows**
 - Any new software installed on the Main Display? Be sure the Display APPs do not litter the CAN-BUS
 - Bus issues can also cause hold-ups, see the orange list above for checking

- **Bus Load % is high**
 - Peaks can happen, if the average is below 60%, it should be fine
 - Were there additional (3rd party) devices installed? Try to remove them and see if the bus load lowers again
 - Constant peaking or average values above 80 may indicate physical issues, and/or a bad device littering the bus that should be removed.
- **Bus works, but can't see message XYZ**
 - If everything else works... the problem lays with the sender. It's either simply not sending that message, or under a difference messageID / Source.